Code No: 126EQ **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, July - 2023
OBJECT ORIENTED ANALYSIS AND DESIGN
(Common to CSE, IT)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART - A

(25 Marks)

1.a)	What is a model? Why is it required during OOAD?	[2]
b)	What are the different phases of OOSDLC? Brief them.	[3]
c)	In what way class diagram is different than object diagram?	[2]
d)	Differentiate between Generalization, Aggregation and Composition.	[3]
e)	What is the purpose of join and fork in activity diagrams?	[2]
f)	Define usecase and explain the relationships between usecases.	[3]
g)	State the differences between event and signal in state diagrams.	[2]
h)	What is the purpose of component diagrams? Explain.	[3]
i)	Write the significance of artifact diagrams.	[2]
j)	Illustrate the differences between patterns and frameworks.	[3]

PART - B

(50 Marks)

2.a) What are adornments in UML? Give suitable examples.
b) Explain the principles of modeling. [5+5]

OR

3.a) Explain the essential features of object oriented modeling.
b) Describe the conceptual model of the UML. [5+5]

4.a) Why is it necessary to have a variety of diagrams in a model of a system?
b) Which UML diagrams give a static view and which give a dynamic view of a system? [5+5]

OR

5.a) What is an interface? How is it represented in UML? Explain.
b) In what way do interfaces help in developing an object oriented system? Discuss. [5+5]

6. What are the common uses of use case diagrams? Consider any two modeling issues and draw relevant use case diagrams. Explain them in detail. [10]

OR

7.a) Describe the use of swim lanes in activity diagrams.
b) Differentiate between sequence and collaboration diagrams. [5+5]

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8.a) Define the State, Event, Transition and give their representation in UML.
b) Enumerate the steps to model family of exceptions. [5+5]

OR

9. Specify the steps to model the following. Illustrate UML diagrams and explain.
a) Modeling processes and devices.
b) Modeling distribution of components. [5+5]

10. What are the packages that can be developed for the unified library system? Explain them in detail. [10]

OR

11. Develop the use case and activity diagrams for unified library application and explain. [10]

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